SHADCN/UI – COMPLETE CHEATSHEET (BEGINNER → ADVANCED)

1. What is shadcn/ui?

- React UI component system
- Built with Tailwind CSS + Radix UI
- Not installed; code is copied into /components/ui
- Fully customizable
- Uses CVA (class-variance-authority)

2. Installation

```
npx shadcn@latest init
npx shadcn@latest add button
```

3. Project Structure

```
components/
ui/
lib/
utils.ts
components.json
```

4. cn() Utility

Merges Tailwind classes safely.

5. Button Component Structure

Includes:

- CVA variants
- forwardRef
- Radix primitive wrapping

6. Using Variants

Click

7. asChild

Allows passing styling to any element.

Dashboard

8. Common Components

- Button, Input, Label, Badge, Card
- Dialog, Tooltip, Popover, Sheet
- Form + RHF + Zod
- Sidebar, Accordion, Tabs
- Table, Calendar, CommandMenu

9. Dialog Example

Open

...

10. Forms

Uses react-hook-form + zod.

11. Dark Mode

Uses CSS variables + ThemeProvider.

12. CVA Variants

```
className={cn(buttonVariants({ variant: "outline" }))}
```

13. Tailwind Utility Styling

Save

14. Radix Primitives

Dialog, Tooltip, DropdownMenu, Popover, etc.

15. Sidebar Pattern

16. Customize Components

Add new variants inside cva() configuration.

17. Best Practices

- Keep UI inside /components/ui
- Use cn()

- Use asChild with Link
- Do not modify Radix primitives

18. Advanced Tips

- Build your own shadcn component
- Create custom design system
- Ideal with Zustand + Server Actions